

YASH RANA

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Profile

Software Developer and recent **Master of Applied Computer Science** graduate from Concordia University with hands-on experience across **backend development, AI/ML, and full-stack web applications**. Skilled in Java, Python, and JavaScript with practical exposure to **LLMs, distributed systems, and cloud platforms**. Co-author of two research publications and known for delivering in deadline-driven environments.

Education

Concordia University <i>Master of Applied Computer Science (MApCompSc)</i>	Sep 2024 – May 2026 <i>Montréal, QC</i>
Gujarat Technological University <i>Bachelor of Engineering in Computer Engineering</i>	Sep 2020 – Apr 2024 <i>Gujarat, India</i>

Work Experience

Blue Data Consulting <i>Data Science Intern</i>	Jul 2023 – Jun 2024 <i>Surat, Gujarat, India</i>
- Built an end-to-end AI content generation system converting client descriptions into structured text, audio, and video outputs using LLMs, LangChain, and Azure Cognitive Services - Engineered a text-to-speech and video generation workflow with Azure Speech + Video Indexer APIs, cutting generation time from 25 → 8 minutes (-68%) at 92% accuracy, while improving generative consistency by 30% via MLflow tracking and prompt refinement	

The Sparks Foundation <i>Data Analyst Intern</i>	Feb 2023 – May 2023 <i>Remote</i>
- Performed exploratory data analysis, predictive modeling, and visualization using Pandas, NumPy, and Scikit-learn to convert raw datasets into actionable business insights - Delivered weekly analytical reports and dashboards summarizing key business metrics, applying statistical analysis and supervised learning techniques to support data-driven decision making	

DevTown <i>Full Stack Developer Intern</i>	Sep 2022 – Feb 2023 <i>Remote</i>
- Delivered a MERN-stack portal for an ed-tech client; active users grew from 120 → 220 (+83%) after introducing interactive dashboards - Built secure React.js + Node.js/Express APIs with JWT auth and optimized routing, reducing backend errors by 45% and improving load speed by 40%	

Research & Projects

Early Detection of Diabetes using Machine Learning <i>Published Paper</i>	Python SVM Scikit-learn
Published in Atlantis Press (ICAA AI 2025), presenting an SVM-based diagnostic model that achieved 95% accuracy for early-stage diabetes prediction, benchmarked via ROC-AUC against Logistic Regression, Random Forest, and KNN	

Redline Markets <i>Source Code</i> <i>Live Demo</i>	Next.js TypeScript Spring Boot Firebase
Built a full-stack F1 prediction market platform with Polymarket-style trading cards, GP Coins economy, Stripe checkout, Firebase Auth, and a Spring Boot REST + WebSocket backend for real-time chat and trades	

Chess Application & Engine <i>Source Code</i>	Python Minimax Alpha-Beta Pruning
Building a chess application and custom engine implementing legal move generation, castling, en passant, and a decision layer using Minimax with Alpha-Beta pruning, iterative deepening, and a custom evaluation function	

Distributed Share Market System (DSMS) <i>Source Code</i>	Java UDP Distributed Systems
Built a fault-tolerant replicated trading system using active replication, consensus, and autonomous recovery to maintain high availability under crash and Byzantine failures	

Online Movie Ticket Booking System <i>Source Code</i> <i>Demo</i>	React Spring Boot MySQL
Developed a full-stack ticket booking platform with authentication, real-time seat selection, simulated secure payments, and role-based access for admin and customer flows	

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, C, HTML5, CSS

Frameworks: FastAPI, Spring Boot, React, Next.js, Node.js, Flask, Express, React Native

Machine Learning & AI: scikit-learn, TensorFlow, XGBoost, LSTM, LangChain, Pandas, NumPy, Matplotlib, Seaborn

Cloud Platforms: AWS (EC2, S3, Lambda), Azure (Cognitive Services, Speech, Video Indexer), Google Cloud

DevOps & CI/CD: Docker, Kubernetes, GitHub Actions, CI/CD Pipelines, Vercel, Netlify, Linux/Unix, Shell Scripting

Databases: MySQL, PostgreSQL, Firebase, Supabase, Prisma, MongoDB

Tools & Expertise: Git, Jira, Postman, PowerBI, MLflow, Distributed Systems, API Design, LLM Integration